

Taylor's Theorem, Inverse/Implicit Functions, and Extrema Review

Lecture 6 for Multivariable Calculus

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1 Lecture Scope and How It Connects to Earlier Lectures

Today we connect several themes that have already appeared separately:

- **Lecture 3** already introduced continuity and the intermediate value theorem on connected domains.
- **Lecture 4** already introduced multivariable chain rules, directional derivatives, Jacobian matrices, and Lagrange multipliers.
- **Lecture 5** already introduced the extreme value theorem, Hessian test, and absolute extrema on closed regions.

So in Lecture 6 we do *not* repeat all of those topics from the beginning. Instead, we use them as tools for three bigger goals:

- understand Taylor expansion as the local polynomial model of a multivariable function,
- understand when a map can be locally inverted and when one variable can be solved from an equation,
- organize extrema problems into a single framework with and without constraints.

Central Theme

The Jacobian gives the best linear approximation. Taylor's theorem improves that to a quadratic approximation. The inverse and implicit function theorems say that when the linear part is nondegenerate, local geometry behaves regularly.

2 Quick Review: Continuity, IVT, Jacobian, Chain Rule

2.1 Intermediate Value Theorem: Already Covered in Lecture 3

For one variable, if h is continuous on $[a, b]$, then every value between $h(a)$ and $h(b)$ is attained.

For multivariable functions, the practical version is:

If $D \subset \mathbb{R}^n$ is connected and $f : D \rightarrow \mathbb{R}$ is continuous, then $f(D)$ is an interval in $\mathbb{R}$.

So if $f(p) < 0 < f(q)$ for two points $p, q \in D$, then there exists some $r \in D$ with $f(r) = 0$.

Why we mention this again. Later, when discussing inverse and implicit functions, we often combine:

- continuity to control sign changes,
- differentiability to control local linear behavior,
- connectedness to avoid “jumping over” values.

2.2 Jacobian Review

For $F = (F_1, \dots, F_m) : \mathbb{R}^n \rightarrow \mathbb{R}^m$, the Jacobian matrix is

$$J_F(a) = \left[\frac{\partial F_i}{\partial x_j}(a) \right].$$

If F is differentiable at a , then

$$F(a + h) = F(a) + J_F(a)h + o(\|h\|).$$

So the Jacobian is the best linear map near a .

Why is it the best linear map? Suppose F is differentiable at a . By definition, there exists a linear map $L : \mathbb{R}^n \rightarrow \mathbb{R}^m$ such that

$$F(a + h) = F(a) + Lh + r(h), \quad \frac{\|r(h)\|}{\|h\|} \rightarrow 0 \quad \text{as } h \rightarrow 0.$$

We claim that this linear map must be $L = J_F(a)$.

Write

$$L = \begin{bmatrix} \ell_{11} & \cdots & \ell_{1n} \\ \vdots & \ddots & \vdots \\ \ell_{m1} & \cdots & \ell_{mn} \end{bmatrix}.$$

Take $h = te_j$, where e_j is the j th standard basis vector. Then

$$\frac{F(a + te_j) - F(a)}{t} = Le_j + \frac{r(te_j)}{t}.$$

As $t \rightarrow 0$, the remainder term tends to 0, so

$$\lim_{t \rightarrow 0} \frac{F(a + te_j) - F(a)}{t} = Le_j.$$

But the left-hand side is exactly the column of partial derivatives

$$\begin{bmatrix} \frac{\partial F_1}{\partial x_j}(a) \\ \vdots \\ \frac{\partial F_m}{\partial x_j}(a) \end{bmatrix}.$$

Therefore the j th column of L equals the j th column of $J_F(a)$ for every j , hence

$$L = J_F(a).$$

So there is only one linear map whose error is of order $o(\|h\|)$, and that is why the Jacobian is the *best* linear approximation near a .

Key preview. If the Jacobian matrix is invertible, then locally the map behaves like an invertible linear transformation. This is exactly the idea behind the inverse function theorem.

2.3 Chain Rule Review

Let $F : \mathbb{R}^n \rightarrow \mathbb{R}^m$ and $G : \mathbb{R}^m \rightarrow \mathbb{R}^k$ be differentiable. Then

$$J_{G \circ F}(a) = J_G(F(a)) J_F(a).$$

This is the matrix form of the chain rule.

Special case 1: scalar-valued function along a curve. If $z = f(x, y)$ and $x = x(t), y = y(t)$, then

$$\frac{dz}{dt} = f_x \frac{dx}{dt} + f_y \frac{dy}{dt}.$$

Special case 2: scalar-valued function of two intermediate variables. If

$$z = f(u, v), \quad u = u(s, t), \quad v = v(s, t),$$

then

$$\frac{\partial z}{\partial s} = f_u \frac{\partial u}{\partial s} + f_v \frac{\partial v}{\partial s}, \quad \frac{\partial z}{\partial t} = f_u \frac{\partial u}{\partial t} + f_v \frac{\partial v}{\partial t}.$$

In Jacobian form,

$$\begin{bmatrix} z_s & z_t \end{bmatrix} = \begin{bmatrix} f_u & f_v \end{bmatrix} \begin{bmatrix} u_s & u_t \\ v_s & v_t \end{bmatrix}.$$

Special case 3: vector-valued map after a parameterization. If

$$F : \mathbb{R}^2 \rightarrow \mathbb{R}^3, \quad (x, y) = (x(s, t), y(s, t)),$$

then

$$J_{F \circ \gamma}(s, t) = J_F(x(s, t), y(s, t)) J_\gamma(s, t),$$

where

$$\gamma(s, t) = (x(s, t), y(s, t)).$$

This is the standard setup for surfaces and coordinate changes.

Special case 4: inverse map. If F is locally invertible, then applying the chain rule to

$$F^{-1} \circ F = \text{Id}$$

gives

$$J_{F^{-1}}(F(a)) J_F(a) = I,$$

so

$$J_{F^{-1}}(F(a)) = (J_F(a))^{-1}.$$

Special case 5: implicit differentiation. If

$$F(x, y(x)) = 0,$$

then the chain rule gives

$$F_x + F_y y'(x) = 0,$$

hence

$$y'(x) = -\frac{F_x}{F_y}$$

whenever $F_y \neq 0$.

2.4 Short Exercises on the Jacobian Form of Chain Rule

1. Let

$$f(u, v) = u^2 + uv, \quad u = s + t, \quad v = s - t.$$

Compute z_s and z_t for $z = f(u(s, t), v(s, t))$ using the chain rule.

2. Let

$$F(x, y) = (x^2 - y, xy, x + y), \quad \gamma(t) = (\cos t, \sin t).$$

Compute the derivative of $F(\gamma(t))$ at general t using Jacobians.

3. Let

$$F(x, y) = (x + y, x - y).$$

Compute J_F , then use the identity

$$J_{F^{-1}}(F(a)) = (J_F(a))^{-1}$$

to find the Jacobian of the inverse map.

4. Suppose

$$G(x, y, z) = x^2 + y^2 + z^2 - 1 = 0$$

defines z locally as a function of x, y . Use the chain rule to derive formulas for z_x and z_y .

Exercise Sketches

- 1.

$$f_u = 2u + v, \quad f_v = u.$$

Also

$$u_s = u_t = 1, \quad v_s = 1, \quad v_t = -1.$$

Therefore

$$z_s = (2u + v) \cdot 1 + u \cdot 1 = 3u + v, \quad z_t = (2u + v) \cdot 1 + u \cdot (-1) = u + v.$$

- 2.

$$J_F(x, y) = \begin{bmatrix} 2x & -1 \\ y & x \\ 1 & 1 \end{bmatrix}, \quad J_\gamma(t) = \begin{bmatrix} -\sin t \\ \cos t \end{bmatrix}.$$

Multiply to get

$$\frac{d}{dt} F(\gamma(t)) = \begin{bmatrix} -2 \cos t \sin t - \cos t \\ -\sin^2 t + \cos^2 t \\ \cos t - \sin t \end{bmatrix}.$$

- 3.

$$J_F = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \quad (J_F)^{-1} = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}.$$

So this is the Jacobian of F^{-1} at every point.

4. Differentiate

$$G(x, y, z(x, y)) = 0$$

with respect to x and y :

$$G_x + G_z z_x = 0, \quad G_y + G_z z_y = 0.$$

Hence

$$z_x = -\frac{G_x}{G_z} = -\frac{x}{z}, \quad z_y = -\frac{G_y}{G_z} = -\frac{y}{z},$$

provided $z \neq 0$.

2.5 Two Chain-Rule Applications: Laplacian in New Coordinates

The chain rule becomes especially important when we rewrite differential operators after a change of variables.

Application 1: A radial function in polar coordinates. Let

$$u(x, y) = f(r), \quad r = \sqrt{x^2 + y^2}.$$

Then

$$u_x = f'(r) \frac{x}{r}, \quad u_y = f'(r) \frac{y}{r}.$$

Differentiate again:

$$u_{xx} = f''(r) \frac{x^2}{r^2} + f'(r) \frac{y^2}{r^3}, \quad u_{yy} = f''(r) \frac{y^2}{r^2} + f'(r) \frac{x^2}{r^3}.$$

So the Laplacian becomes

$$\Delta u = u_{xx} + u_{yy} = f''(r) + \frac{1}{r} f'(r).$$

Meaning. If a function depends only on the distance r from the origin, then the two-variable operator Δ reduces to a one-variable expression.

Application 2: Polar-coordinate form of the Laplacian. Now let

$$u(x, y) = v(r, \theta), \quad x = r \cos \theta, \quad y = r \sin \theta.$$

Using the chain rule,

$$u_x = v_r r_x + v_\theta \theta_x, \quad u_y = v_r r_y + v_\theta \theta_y.$$

Since

$$r_x = \frac{x}{r} = \cos \theta, \quad r_y = \frac{y}{r} = \sin \theta,$$

and

$$\theta_x = -\frac{y}{r^2} = -\frac{\sin \theta}{r}, \quad \theta_y = \frac{x}{r^2} = \frac{\cos \theta}{r},$$

we get

$$u_x = v_r \cos \theta - \frac{1}{r} v_\theta \sin \theta, \quad u_y = v_r \sin \theta + \frac{1}{r} v_\theta \cos \theta.$$

Differentiating once more and simplifying gives the standard formula

$$\Delta u = u_{xx} + u_{yy} = v_{rr} + \frac{1}{r} v_r + \frac{1}{r^2} v_{\theta\theta}.$$

Why this is useful. This formula is fundamental in solving PDEs with circular symmetry, such as heat flow or potential problems on disks.

3 Taylor's Theorem for Multivariable Functions

3.1 Recall: Differentiability at a Point

Before writing Taylor expansions, recall the definition of differentiability.

Let $F : \mathbb{R}^n \rightarrow \mathbb{R}^m$ and let $a \in \mathbb{R}^n$. We say that F is **differentiable at** a if there exists a linear map $L : \mathbb{R}^n \rightarrow \mathbb{R}^m$ such that

$$F(a+h) = F(a) + Lh + r(h), \quad \frac{\|r(h)\|}{\|h\|} \rightarrow 0 \quad \text{as } h \rightarrow 0.$$

That is, near a , the function is approximated by a linear map, and the error is small compared with $\|h\|$.

When $F = f : \mathbb{R}^n \rightarrow \mathbb{R}$ is scalar-valued, this becomes

$$f(a+h) = f(a) + \nabla f(a) \cdot h + o(\|h\|).$$

So differentiability is exactly the statement that the first-order Taylor formula is valid.

3.2 First-Order Taylor Formula

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be differentiable at a . Then

$$f(a+h) = f(a) + \nabla f(a) \cdot h + o(\|h\|).$$

This is the linear approximation already used in tangent-plane formulas.

3.3 Second-Order Taylor Formula

Assume $f \in C^2$ near $a \in \mathbb{R}^n$. Then

$$f(a+h) = f(a) + \nabla f(a) \cdot h + \frac{1}{2} h^\top H_f(a) h + o(\|h\|^2),$$

where $H_f(a)$ is the Hessian matrix

$$H_f(a) = \left[\frac{\partial^2 f}{\partial x_i \partial x_j}(a) \right].$$

In two variables, with $a = (x_0, y_0)$ and $h = (\Delta x, \Delta y)$,

$$\begin{aligned} f(x_0 + \Delta x, y_0 + \Delta y) &= f(x_0, y_0) + f_x(x_0, y_0)\Delta x + f_y(x_0, y_0)\Delta y \\ &\quad + \frac{1}{2} \left(f_{xx}(\Delta x)^2 + 2f_{xy}\Delta x\Delta y + f_{yy}(\Delta y)^2 \right) + o((\Delta x)^2 + (\Delta y)^2). \end{aligned}$$

Pitfall 1

In multivariable Taylor expansion, the quadratic term is *one whole quadratic form*. Students often remember f_{xx}, f_{xy}, f_{yy} separately but forget the compact matrix structure

$$\frac{1}{2} h^\top H_f(a) h.$$

3.4 Example 1: Quadratic Approximation

Let

$$f(x, y) = e^{x+y}.$$

At $(0, 0)$:

$$f(0, 0) = 1, \quad f_x(0, 0) = 1, \quad f_y(0, 0) = 1,$$

and

$$f_{xx}(0, 0) = f_{xy}(0, 0) = f_{yy}(0, 0) = 1.$$

So near $(0, 0)$,

$$f(x, y) \approx 1 + x + y + \frac{1}{2}(x^2 + 2xy + y^2) = 1 + x + y + \frac{1}{2}(x + y)^2.$$

3.5 Higher-Order Taylor Expansion with Peano Remainder

More generally, if $f \in C^m$ near $a \in \mathbb{R}^n$, then

$$f(a + h) = \sum_{|\alpha| \leq m} \frac{D^\alpha f(a)}{\alpha!} h^\alpha + o(\|h\|^m) \quad (h \rightarrow 0).$$

This is the multivariable Taylor formula with **Peano remainder**.

Multi-index notation. For

$$\alpha = (\alpha_1, \dots, \alpha_n), \quad |\alpha| = \alpha_1 + \dots + \alpha_n,$$

we write

$$\alpha! = \alpha_1! \dots \alpha_n!, \quad h^\alpha = h_1^{\alpha_1} \dots h_n^{\alpha_n}, \quad D^\alpha f = \frac{\partial^{|\alpha|} f}{\partial x_1^{\alpha_1} \dots \partial x_n^{\alpha_n}}.$$

Meaning of the Peano remainder. The term

$$o(\|h\|^m)$$

means that after subtracting the Taylor polynomial of degree m , the remaining error is negligible compared with $\|h\|^m$.

Two-variable form up to degree three. If $f \in C^3$ near (x_0, y_0) and

$$h = (\Delta x, \Delta y),$$

then

$$\begin{aligned} f(x_0 + \Delta x, y_0 + \Delta y) &= f(x_0, y_0) + f_x \Delta x + f_y \Delta y \\ &\quad + \frac{1}{2} \left(f_{xx} (\Delta x)^2 + 2f_{xy} \Delta x \Delta y + f_{yy} (\Delta y)^2 \right) \\ &\quad + \frac{1}{6} \left(f_{xxx} (\Delta x)^3 + 3f_{xxy} (\Delta x)^2 \Delta y + 3f_{xyy} \Delta x (\Delta y)^2 + f_{yyy} (\Delta y)^3 \right) \\ &\quad + o\left(\left((\Delta x)^2 + (\Delta y)^2\right)^{3/2}\right), \end{aligned}$$

where all derivatives on the right are evaluated at (x_0, y_0) .

Why this matters. Sometimes the quadratic part vanishes or is inconclusive. Then higher-order Taylor terms help determine local behavior.

Example 1A. Let

$$f(x, y) = \sin(x + y).$$

At $(0, 0)$, using $\sin u = u - \frac{u^3}{6} + o(u^3)$ with $u = x + y$, we get

$$f(x, y) = x + y - \frac{(x + y)^3}{6} + o((x^2 + y^2)^{3/2}).$$

So the third-order Taylor polynomial is

$$T_3(x, y) = x + y - \frac{1}{6}(x + y)^3.$$

3.6 Why Taylor's Theorem Matters for Extrema

If a is a critical point, then $\nabla f(a) = 0$, so the first-order term disappears:

$$f(a + h) - f(a) = \frac{1}{2} h^\top H_f(a) h + o(\|h\|^2).$$

Therefore the local behavior is controlled by the Hessian quadratic form.

Interpretation.

- positive definite Hessian $\Rightarrow$ local minimum,
- negative definite Hessian $\Rightarrow$ local maximum,
- indefinite Hessian $\Rightarrow$ saddle point.

3.7 Example 2: Recovering the Second Derivative Test

Let

$$f(x, y) = x^2 + xy + y^2.$$

At $(0, 0)$,

$$\nabla f(0, 0) = (0, 0), \quad H_f(0, 0) = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}.$$

Then

$$h^\top H_f(0, 0) h = 2h_1^2 + 2h_1h_2 + 2h_2^2 = (h_1 + h_2)^2 + h_1^2 + h_2^2 > 0$$

for $h \neq 0$. Hence $(0, 0)$ is a strict local minimum.

4 From Linear Algebra to Local Invertibility

4.1 The Guiding Linear Model

For a linear map $L : \mathbb{R}^n \rightarrow \mathbb{R}^n$:

- if $\det L \neq 0$, then L is invertible;
- if $\det L = 0$, then L collapses some direction and cannot be inverted.

The inverse function theorem says that a differentiable nonlinear map behaves the same way *locally*.

4.2 Inverse Function Theorem

Theorem 4.1 (Inverse Function Theorem, local version). *Let $F : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be C^1 in a neighborhood of a . If*

$$\det J_F(a) \neq 0,$$

then there exist neighborhoods U of a and V of $F(a)$ such that

- *$F : U \rightarrow V$ is one-to-one and onto,*
- *the inverse map $F^{-1} : V \rightarrow U$ exists,*
- *F^{-1} is C^1 .*

Proof idea. Because $\det J_F(a) \neq 0$, the Jacobian matrix $J_F(a)$ is invertible. So the linearization

$$F(a + h) \approx F(a) + J_F(a)h$$

behaves like a bijective linear map near $h = 0$.

Set

$$b = F(a), \quad A = J_F(a).$$

Since F is differentiable at a ,

$$F(a + h) = b + Ah + r(h), \quad \frac{\|r(h)\|}{\|h\|} \rightarrow 0.$$

Thus for sufficiently small h , the nonlinear error $r(h)$ is much smaller than the main linear term Ah .

Step 1: local injectivity. Suppose $F(a + h_1) = F(a + h_2)$ for two nearby points. Then

$$A(h_1 - h_2) = -(r(h_1) - r(h_2)).$$

Because A is invertible, the left-hand side controls $\|h_1 - h_2\|$ from below, while differentiability makes the right-hand side small compared with $\|h_1 - h_2\|$ when h_1, h_2 are close to 0. This is only possible if

$$h_1 = h_2.$$

So F is one-to-one near a .

Step 2: local surjectivity onto a neighborhood of b . Since A is invertible, for every y near b the linear equation

$$b + Ah = y$$

has a unique solution

$$h = A^{-1}(y - b).$$

The true nonlinear equation

$$F(a + h) = y$$

is therefore a small perturbation of a solvable linear equation. One can rewrite it as a fixed-point problem

$$h = A^{-1}(y - b - r(h)),$$

and for y near b this map sends a small ball into itself and is a contraction. Hence it has a unique fixed point. Therefore every y in some neighborhood of b has a unique preimage near a .

Step 3: differentiability of the inverse. Now let x be near a and $y = F(x)$. Since

$$F^{-1}(F(x)) = x,$$

the chain rule gives

$$J_{F^{-1}}(F(x)) J_F(x) = I.$$

Hence

$$J_{F^{-1}}(F(x)) = (J_F(x))^{-1}.$$

Because F is C^1 , the matrix $J_F(x)$ depends continuously on x , and inversion is continuous on invertible matrices. Therefore F^{-1} is also C^1 . $\square$

Interpretation. Near a , the map F can be solved uniquely for the input in terms of the output.

Derivative of the inverse. If $b = F(a)$, then

$$J_{F^{-1}}(b) = (J_F(a))^{-1}.$$

This follows from the chain rule applied to

$$F^{-1} \circ F = \text{Id}, \quad F \circ F^{-1} = \text{Id}.$$

4.3 Example 3: A Local Inverse Exists

Let

$$F(x, y) = (e^x \cos y, e^x \sin y).$$

Then

$$J_F(x, y) = \begin{bmatrix} e^x \cos y & -e^x \sin y \\ e^x \sin y & e^x \cos y \end{bmatrix}, \quad \det J_F(x, y) = e^{2x} > 0.$$

So at every point, the Jacobian determinant is nonzero. Therefore F is locally invertible everywhere.

Important caution. Locally invertible does *not* necessarily mean globally invertible. In this example, periodicity in y prevents global one-to-one behavior.

5 Implicit Function Theorem

5.1 Basic Two-Variable Version

Suppose an equation

$$F(x, y) = 0$$

defines y implicitly as a function of x near a point (a, b) .

Theorem 5.1 (Implicit Function Theorem, two-variable version). *Let $F(x, y)$ be C^1 near (a, b) and assume*

$$F(a, b) = 0, \quad F_y(a, b) \neq 0.$$

Then there exists a neighborhood of a and a unique C^1 function $y = \varphi(x)$ such that

$$\varphi(a) = b, \quad F(x, \varphi(x)) = 0$$

for all nearby x .

Proof idea. Define a new map

$$G(x, y) = (x, F(x, y)).$$

Then

$$G(a, b) = (a, 0).$$

Its Jacobian matrix is

$$J_G(x, y) = \begin{bmatrix} 1 & 0 \\ F_x(x, y) & F_y(x, y) \end{bmatrix},$$

so at (a, b) ,

$$\det J_G(a, b) = F_y(a, b) \neq 0.$$

Therefore, by the inverse function theorem, G is locally invertible near (a, b) .

This means that for (u, v) near $(a, 0)$ there is a unique nearby pair (x, y) such that

$$G(x, y) = (u, v).$$

In particular, if we take $v = 0$, then for each u near a there is a unique nearby (x, y) satisfying

$$G(x, y) = (u, 0).$$

But the first component of $G(x, y)$ is just x , so necessarily

$$x = u.$$

The second component then gives

$$F(u, y) = 0.$$

Thus for each u near a , there is a unique nearby $y = \varphi(u)$ such that

$$F(u, \varphi(u)) = 0.$$

Also, because G^{-1} is C^1 , this function φ is C^1 . Finally, since $F(a, b) = 0$, we have

$$\varphi(a) = b.$$

Hence the required implicit function exists and is unique near (a, b) . □

Meaning. If the equation is nondegenerate in the y -direction, then near (a, b) we can solve for y as a function of x .

5.2 Derivative Formula

If $F(x, \varphi(x)) = 0$, then by the chain rule,

$$F_x(x, \varphi(x)) + F_y(x, \varphi(x))\varphi'(x) = 0.$$

Hence

$$\varphi'(x) = -\frac{F_x(x, \varphi(x))}{F_y(x, \varphi(x))}.$$

At the point $x = a$:

$$\varphi'(a) = -\frac{F_x(a, b)}{F_y(a, b)}.$$

Worked problem: computing the derivative of an implicitly defined function. Suppose

$$x^2 + xy + y^2 = 3$$

defines y as a function of x near the point $(1, 1)$. Compute $y'(1)$.

Solution. Let

$$F(x, y) = x^2 + xy + y^2 - 3.$$

Then

$$F(1, 1) = 1 + 1 + 1 - 3 = 0.$$

Also

$$F_y(x, y) = x + 2y, \quad F_y(1, 1) = 3 \neq 0.$$

So the implicit function theorem applies, and near $(1, 1)$ the equation defines a differentiable function $y = \varphi(x)$.

Now compute

$$F_x(x, y) = 2x + y.$$

Therefore

$$\varphi'(1) = -\frac{F_x(1, 1)}{F_y(1, 1)} = -\frac{2 \cdot 1 + 1}{1 + 2 \cdot 1} = -\frac{3}{3} = -1.$$

Hence

$$y'(1) = -1.$$

5.3 Example 4: Circle as an Implicit Curve

Let

$$F(x, y) = x^2 + y^2 - 1.$$

At the point

$$\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right),$$

we have

$$F = 0, \quad F_y = 2y = \sqrt{3} \neq 0.$$

So near this point, the circle can be written as a function $y = \varphi(x)$.

Its derivative is

$$\varphi'(x) = -\frac{F_x}{F_y} = -\frac{2x}{2y} = -\frac{x}{y}.$$

At $\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$:

$$\varphi'\left(\frac{1}{2}\right) = -\frac{1/2}{\sqrt{3}/2} = -\frac{1}{\sqrt{3}}.$$

5.4 Why the Implicit Function Theorem Is Related to the Inverse Function Theorem

The implicit function theorem can be viewed as a consequence of the inverse function theorem applied to a carefully chosen map.

Idea. If we define

$$G(x, y) = (x, F(x, y)),$$

then

$$J_G(a, b) = \begin{bmatrix} 1 & 0 \\ F_x(a, b) & F_y(a, b) \end{bmatrix}, \quad \det J_G(a, b) = F_y(a, b).$$

So if $F_y(a, b) \neq 0$, then G is locally invertible.

What does it mean to invert G ? By the inverse function theorem, there exist neighborhoods U of (a, b) and V of $(a, 0)$ such that

$$G : U \rightarrow V$$

has an inverse

$$G^{-1} : V \rightarrow U.$$

Write

$$G^{-1}(u, v) = (X(u, v), Y(u, v)).$$

This means:

$$G(x, y) = (u, v) \iff (x, y) = (X(u, v), Y(u, v)).$$

Since

$$G(x, y) = (x, F(x, y)),$$

the equation

$$G(x, y) = (u, v)$$

is exactly the system

$$x = u, \quad F(x, y) = v.$$

So after inversion, $X(u, v)$ and $Y(u, v)$ tell us how to recover the original variables (x, y) from the new variables (u, v) .

Why this solves the implicit equation. To solve

$$F(x, y) = 0$$

near (a, b) , we set

$$v = 0.$$

Then

$$G(x, y) = (u, 0)$$

means

$$x = u, \quad F(x, y) = 0.$$

Applying the inverse map gives

$$(x, y) = G^{-1}(u, 0) = (X(u, 0), Y(u, 0)).$$

But the first component must satisfy $x = u$, so in fact

$$X(u, 0) = u.$$

Therefore the second component gives the desired implicit function:

$$y = Y(u, 0).$$

If we define

$$\varphi(u) = Y(u, 0),$$

then

$$F(u, \varphi(u)) = 0.$$

This is the local solution guaranteed by the implicit function theorem.

Summary of the mechanism. The inverse function theorem solves the two equations

$$u = x, \quad v = F(x, y)$$

for (x, y) in terms of (u, v) . The implicit function theorem is the special case obtained by freezing

$$v = 0.$$

So inverting G lets us solve the level set equation $F(x, y) = 0$ for one variable as a function of the other.

6 Extrema Without Constraints and With Constraints

6.1 Unconstrained Extrema: The Interior Picture

Suppose $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is differentiable.

Necessary condition. If a is an interior local extremum, then

$$\nabla f(a) = 0.$$

Second-order test. If $\nabla f(a) = 0$, then Taylor's theorem gives

$$f(a + h) - f(a) = \frac{1}{2} h^\top H_f(a) h + o(\|h\|^2).$$

To interpret the Hessian, recall the following matrix definitions for a symmetric matrix A :

- A is **positive definite** if

$$h^\top A h > 0 \quad \text{for every } h \neq 0.$$

- A is **negative definite** if

$$h^\top A h < 0 \quad \text{for every } h \neq 0.$$

- A is **indefinite** if the quadratic form $h^\top A h$ takes both positive and negative values for different nonzero vectors h .

Useful criterion for a 2×2 symmetric matrix. Let

$$A = \begin{bmatrix} a & b \\ b & c \end{bmatrix}.$$

Then:

- A is positive definite if and only if

$$a > 0, \quad \det A = ac - b^2 > 0.$$

- A is negative definite if and only if

$$a < 0, \quad \det A = ac - b^2 > 0.$$

- A is indefinite if

$$\det A = ac - b^2 < 0.$$

This is exactly the criterion used in the two-variable second derivative test.

So:

- Hessian positive definite $\Rightarrow$ local min,
- Hessian negative definite $\Rightarrow$ local max,
- Hessian indefinite $\Rightarrow$ saddle.

6.2 Constrained Extrema: The Geometric Picture

Now suppose we optimize $f(x, y)$ subject to

$$g(x, y) = c.$$

At a regular point of the constraint curve ($\nabla g \neq 0$), the feasible motions are tangent to the curve.

At a constrained extremum, the first-order change in every feasible tangent direction must vanish. So ∇f is orthogonal to the tangent direction and therefore parallel to ∇g :

$$\nabla f = \lambda \nabla g.$$

Connection with the implicit function theorem. If $g_y(a, b) \neq 0$, then near (a, b) the constraint can be written as

$$y = \varphi(x).$$

Then the constrained problem becomes a one-variable problem:

$$\Phi(x) = f(x, \varphi(x)).$$

At an optimum,

$$\Phi'(x) = 0.$$

Using the chain rule gives the same condition as Lagrange multipliers.

6.3 Example 5: Constrained Optimization via Implicit View

Optimize

$$f(x, y) = x + y$$

subject to

$$g(x, y) = x^2 + y^2 = 1.$$

Lagrange multiplier method.

$$\nabla f = (1, 1), \quad \nabla g = (2x, 2y).$$

So

$$(1, 1) = \lambda(2x, 2y), \quad x^2 + y^2 = 1.$$

Thus $x = y$, and the candidates are

$$\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right), \quad \left(-\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right).$$

Hence

$$f_{\max} = \sqrt{2}, \quad f_{\min} = -\sqrt{2}.$$

Implicit-function viewpoint. On the upper semicircle, we can write

$$y = \sqrt{1 - x^2}.$$

Then

$$\Phi(x) = x + \sqrt{1 - x^2}.$$

Differentiating yields the same candidate equation.

6.4 Lagrange Multipliers with Multiple Constraints

The same idea works when there is more than one smooth equality constraint.

General form. Suppose we want to optimize $f : \mathbb{R}^n \rightarrow \mathbb{R}$ subject to

$$g_1(x) = c_1, \quad g_2(x) = c_2, \quad \dots, \quad g_k(x) = c_k.$$

At a regular constrained extremum, the feasible directions are tangent to the intersection of these constraint surfaces. Therefore ∇f must be orthogonal to every feasible direction, so it lies in the normal space spanned by the constraint gradients:

$$\nabla f = \lambda_1 \nabla g_1 + \lambda_2 \nabla g_2 + \dots + \lambda_k \nabla g_k.$$

Together with the constraint equations

$$g_1(x) = c_1, \quad \dots, \quad g_k(x) = c_k,$$

this gives a system of equations for the point x and the multipliers $\lambda_1, \dots, \lambda_k$.

How to deal with it in practice.

1. Write down all constraint equations.
2. Compute $\nabla f, \nabla g_1, \dots, \nabla g_k$.
3. Solve

$$\nabla f = \lambda_1 \nabla g_1 + \dots + \lambda_k \nabla g_k$$

together with the constraints.

4. Evaluate f at all candidate points and compare values.

Geometric meaning. With one constraint, ∇f is parallel to ∇g . With several constraints, ∇f does not usually have to be parallel to any single constraint gradient; instead, it must lie in the span of all of them.

Example 5A. Find the extrema of

$$f(x, y, z) = x + y + z$$

subject to

$$x^2 + y^2 + z^2 = 1, \quad x + y = 0.$$

Solution. Let

$$g_1(x, y, z) = x^2 + y^2 + z^2, \quad g_2(x, y, z) = x + y.$$

Then

$$\nabla f = (1, 1, 1), \quad \nabla g_1 = (2x, 2y, 2z), \quad \nabla g_2 = (1, 1, 0).$$

The multiple-constraint Lagrange system is

$$(1, 1, 1) = \lambda(2x, 2y, 2z) + \mu(1, 1, 0),$$

with

$$x^2 + y^2 + z^2 = 1, \quad x + y = 0.$$

Comparing components:

$$1 = 2\lambda x + \mu, \quad 1 = 2\lambda y + \mu, \quad 1 = 2\lambda z.$$

Subtracting the first two equations gives

$$2\lambda(x - y) = 0.$$

Since the third equation implies $\lambda \neq 0$, we get

$$x = y.$$

Together with the constraint $x + y = 0$, this gives

$$x = y = 0.$$

Then from $x^2 + y^2 + z^2 = 1$,

$$z = \pm 1.$$

So the candidate points are

$$(0, 0, 1), \quad (0, 0, -1).$$

Their function values are

$$f(0, 0, 1) = 1, \quad f(0, 0, -1) = -1.$$

Hence

$$f_{\max} = 1, \quad f_{\min} = -1.$$

Remark. If the constraint gradients become linearly dependent or one of them vanishes, the point is not regular, and extra care is needed. In that case, parameterization or direct analysis of the constraint set may be easier.

6.5 Existence of Absolute Extrema

This topic was already proved in Lecture 5, but it fits naturally here:

Theorem 6.1 (Extreme Value Theorem). *If $D \subset \mathbb{R}^n$ is closed and bounded and $f : D \rightarrow \mathbb{R}$ is continuous, then f attains both an absolute maximum and an absolute minimum on D .*

Why it matters in Lecture 6.

- Taylor's theorem helps classify interior candidates.
- The implicit-function/Lagrange viewpoint helps analyze smooth constraints.
- EVT guarantees that on compact domains, the global answers actually exist.

Pitfall 2

Do not mix up three different questions:

- **Does an extremum exist?** Use continuity + compactness.
- **Where are the candidates?** Use $\nabla f = 0$, boundary analysis, or $\nabla f = \lambda \nabla g$.
- **What type is each candidate?** Use Taylor/Hessian or direct comparison.

7 A Compact Workflow for Students

1. Check what type of problem you have: unconstrained, closed-region, or smooth constraint.
2. For local analysis, identify the linear part using the Jacobian/gradient.
3. For second-order analysis, use Taylor expansion and the Hessian.
4. If solving an equation $F(x, y) = 0$, check whether a partial derivative such as F_y is nonzero.
5. If the Jacobian determinant is nonzero, expect local invertibility.
6. For global extrema, compare all candidates and use EVT when the domain is compact.

8 Practice Set

1. Find the second-order Taylor polynomial of

$$f(x, y) = \ln(1 + x + y)$$

at $(0, 0)$.

2. Let

$$F(x, y) = (x + y, x - y).$$

Compute J_F , check whether F is locally invertible, and find the inverse map explicitly.

3. For

$$F(x, y) = x^2 + xy + y^2 - 3,$$

show that near $(1, 1)$ the equation $F(x, y) = 0$ defines y as a function of x , and compute $y'(1)$.

4. Classify the critical point of

$$f(x, y) = x^2 - 4xy + 5y^2.$$

5. Find the constrained extrema of

$$f(x, y) = x^2 + y^2$$

subject to

$$x + y = 1.$$

Solution Sketches

1. Use

$$\ln(1 + u) = u - \frac{1}{2}u^2 + o(u^2), \quad u = x + y,$$

so

$$f(x, y) = x + y - \frac{1}{2}(x + y)^2 + o(x^2 + y^2).$$

- 2.

$$J_F = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \quad \det J_F = -2 \neq 0.$$

Hence F is locally invertible (in fact globally invertible), and

$$F^{-1}(u, v) = \left(\frac{u + v}{2}, \frac{u - v}{2} \right).$$

3. Here

$$F_y(x, y) = x + 2y,$$

so

$$F_y(1, 1) = 3 \neq 0.$$

Thus the implicit function theorem applies, and

$$y'(1) = -\frac{F_x(1, 1)}{F_y(1, 1)} = -\frac{2x + y}{x + 2y} \Big|_{(1, 1)} = -1.$$

4. The only critical point is $(0, 0)$, and

$$H_f = \begin{bmatrix} 2 & -4 \\ -4 & 10 \end{bmatrix}.$$

Its determinant is $20 - 16 = 4 > 0$ and $f_{xx} = 2 > 0$, so $(0, 0)$ is a strict local minimum.

5. Use either substitution $y = 1 - x$ or Lagrange multipliers. The minimum is attained at

$$\left(\frac{1}{2}, \frac{1}{2}\right),$$

with value $\frac{1}{2}$. There is no maximum on the whole line $x + y = 1$ if the constraint were not closed, but on this affine constraint in $\mathbb{R}^2$ the function still goes to infinity, so only the minimum exists.

9 Project-Style Problem

Project question. Suppose $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ is smooth and (a, b) is a critical point:

$$\nabla f(a, b) = 0.$$

Discuss how to determine the local behavior of f near (a, b) when the Hessian matrix

$$H_f(a, b)$$

is either

- **indefinite**, or
- **degenerate** (for example, $\det H_f(a, b) = 0$).

In particular, explain how one can still decide whether (a, b) is a local maximum, local minimum, saddle point, or inconclusive case.

Discussion guide / solution sketch.

1. If the Hessian is **indefinite**, then the quadratic form

$$Q(h) = h^\top H_f(a, b)h$$

takes both positive and negative values. Since

$$f(a + h) - f(a) = \frac{1}{2}Q(h) + o(\|h\|^2),$$

the function increases in some directions and decreases in others. Therefore (a, b) is a **saddle point**.

2. If the Hessian is **positive definite** or **negative definite**, the answer is easy: local minimum or local maximum, respectively. The interesting difficulty is the **degenerate** case.

3. If the Hessian is **degenerate**, then the quadratic term is not enough to decide the sign of

$$f(a+h) - f(a).$$

In this case, one should look at **higher-order Taylor terms**.

4. A standard higher-order strategy is to use the Taylor expansion in multi-index form. Since (a, b) is a critical point, the first-order term vanishes, so if the quadratic term is degenerate we continue until the first nonzero homogeneous term appears:

$$f((a, b) + h) - f(a, b) = \sum_{|\alpha|=m} \frac{D^\alpha f(a, b)}{\alpha!} h^\alpha + o(\|h\|^m),$$

where $m \geq 2$ is the smallest order for which some derivative $D^\alpha f(a, b)$ with $|\alpha| = m$ is nonzero. Denote

$$P_m(h) = \sum_{|\alpha|=m} \frac{D^\alpha f(a, b)}{\alpha!} h^\alpha.$$

Then P_m is a homogeneous polynomial of degree m , and near the critical point the sign of

$$f((a, b) + h) - f(a, b)$$

is determined by the sign of $P_m(h)$.

Case 1: m is odd. Because P_m is homogeneous of odd degree,

$$P_m(-h) = (-1)^m P_m(h) = -P_m(h).$$

So if $P_m(h_0) \neq 0$ for some h_0 , then

$$P_m(h_0) \quad \text{and} \quad P_m(-h_0)$$

have opposite signs. Hence for small $t > 0$,

$$f((a, b) + th_0) - f(a, b)$$

and

$$f((a, b) - th_0) - f(a, b)$$

have opposite signs. Therefore (a, b) is a **saddle point**.

Case 2: m is even and $P_m(h) > 0$ for all $h \neq 0$. Then by homogeneity,

$$P_m(th) = t^m P_m(h) > 0 \quad (t \neq 0).$$

Since the remainder is $o(\|h\|^m)$, for sufficiently small nonzero h it cannot change the sign of the leading term. Thus

$$f((a, b) + h) - f(a, b) > 0$$

for all sufficiently small $h \neq 0$, so (a, b) is a **local minimum**.

Case 3: m is even and $P_m(h) < 0$ for all $h \neq 0$. Exactly the same argument gives

$$f((a, b) + h) - f(a, b) < 0$$

for all sufficiently small $h \neq 0$. Hence (a, b) is a **local maximum**.

Case 4: P_m takes both positive and negative values. Suppose there exist $h_1, h_2 \neq 0$ such that

$$P_m(h_1) > 0, \quad P_m(h_2) < 0.$$

Then for sufficiently small $t > 0$,

$$f((a, b) + th_1) - f(a, b) > 0, \quad f((a, b) + th_2) - f(a, b) < 0.$$

So (a, b) is a **saddle point**.

Case 5: $P_m(h) \geq 0$ or $P_m(h) \leq 0$, but vanishes for some nonzero h . This is the delicate case. Along directions where

$$P_m(h) = 0,$$

the leading nonzero term disappears, so the sign may be decided only by even higher-order terms. Therefore the conclusion is **not automatic**; one must continue the Taylor expansion further or test special paths.

5. Another effective method is to test the function along carefully chosen curves through the critical point, such as

$$y = kx, \quad y = x^2, \quad y = mx^p,$$

or other paths adapted to the formula of f . If along one path $f - f(a, b) > 0$ and along another $f - f(a, b) < 0$, then (a, b) is a saddle point.

6. Typical examples:

•

$$f(x, y) = x^4 + y^4$$

has degenerate Hessian at $(0, 0)$, but $f \geq 0$ and equals 0 only at the origin, so $(0, 0)$ is a local minimum.

•

$$f(x, y) = x^4 - y^4$$

also has degenerate Hessian at $(0, 0)$, but takes both positive and negative values near the origin, so $(0, 0)$ is a saddle point.

•

$$f(x, y) = x^3 + y^2$$

has first nonzero term of odd degree, so sign changes occur and $(0, 0)$ is a saddle point.

7. Final lesson: the Hessian test is powerful, but when it is degenerate the correct tool is **higher-order Taylor expansion plus directional/path analysis**.

10 Summary

- Taylor's theorem upgrades linear approximation to quadratic approximation.
- At a critical point, the Hessian controls the second-order local behavior.
- A nonzero Jacobian determinant means the map is locally invertible.
- A nonzero partial derivative such as F_y lets us solve an equation locally for one variable.
- Constrained extrema can be understood by either implicit-function reduction or Lagrange multipliers.
- IVT, chain rule, Jacobian, and EVT are not isolated facts; they work together in local and global analysis.